SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING COMPUTER SCIENCE AND ENGINEERING LIST OF EXPERIMENTS

SUB CODE & NAME: CSA1712-ARTIFICIAL INTELLIGENCE

3.Write the python program for Water Jug Problem

```
from collections import deque

queue = deque([((0, 0), [])])
visited = {(0, 0)}

while queue:

    (a, b), path = queue.popleft()

    if a == 2 or b == 2:
        print("Solution:")
        for state in path:
            print(state)
        print((a, b))
        break

    states = [
        ((4, b), "Fill 4L jug"),
        ((a, 3), "Fill 3L jug"),
        ((0, b), "Empty 4L jug"),
        ((a, 0), "Empty 3L jug")
    ]

    # Pour 4L into 3L
    amount = min(a, 3 - b)
    states.append(
        ((a - amount, b + amount),
         "Pour 4L into 3L")
    )

    # Pour 3L into 4L
    amount = min(b, 4 - a)
    states.append(
        ((a + amount, b - amount),
         "Pour 3L into 4L")
    )

    for state, action in states:

        if state not in visited:

            visited.add(state)

            queue.append(
                (state, path + [(state, action)])
            )
```

Output

```
== RESTART: C:/Users/Lekha/AppData/Lc
Solution:
((0, 3), 'Fill 3L jug')
((3, 0), 'Pour 3L into 4L')
((3, 3), 'Fill 3L jug')
((4, 2), 'Pour 3L into 4L')
(4, 2)
```